

UNIVERSITY OF BRISTOL

School of Mathematics

FINANCIAL MATHEMATICS

MATH35400

(Paper code MATH-35400)

MAY/JUNE 2021 2 hours 30 minutes

This paper contains 4 questions. All answers will be used for assessment.

There is a table on p. 6 with probabilities of a standard normal distribution that you may find helpful, but you can also use a calculator or computer for calculating those probabilities.

On this examination, the marking scheme is indicative and is intended only as a guide to the relative weighting of the questions.

1. (25 marks)

Stocks of telecommunication company *PlinTo* are currently traded at $S_0 = £8$ per share. A financial company wants to price a European call option on *PlinTo* shares with strike price $K = £5$ maturing in exactly one year. They have access to a bank account with an annual interest rate $r = 0.125$.

- (a) Alex suggests to model the stock by assuming that the stock price in a year's time S_1 is either £11 with probability 0.6 or £3 with probability 0.4. Alex suggests to work out the expected value of the call option and use that as the time 0 price for the option.
 - i. (1 marks) What price would Alex charge?
 - ii. (2 marks) Is the financial market model that Alex suggests free of arbitrage? Give a brief explanation.
 - iii. (3 marks) Find a replicating portfolio for the European call option and determine its time 0 value C_0 and compare with the price that Alex calculated in (i).
 - iv. (2 marks) What would an arbitrageur do at time 0 if the company would sell the option for Alex' price?
- (b) Kim has doubts about Alex' model and is concerned that the stock price might increase to $S_1 = £20$.
 - i. (2 marks) Suppose the company sells one European call option for C_0 from (a) iii. and builds up the replicating portfolio from (a) iii. at time 0. What would be the loss at time 1 if the stock price would actually increase to £20?
 - ii. (7 marks) Kim suggests to consider a model with three states $\omega_i, i = 1, 2, 3$ where $S_1(\omega_1) = £20$, $S_1(\omega_2) = £11$ and $S_1(\omega_3) = £3$ and where $\mathbb{P}(\{\omega_i\}) > 0$ for each i . Determine all risk-neutral probability measures. Show that the European Call option is not attainable under Kim's model?
- (c) Chris suggests that a two-period model with the same possible stock prices as in Kim's model at time $T = 2$ might work well.
 - i. (2 marks) Give without explanation an example for a price process $S_t(\omega), t = 0, 1, 2, \omega \in \Omega$ such that the two-period model is complete and that S_2 takes values £7, £13 and £20.
 - ii. (6 marks) Calculate the risk-neutral time 0 price of the European Call option under your model.
[Hint: You may need to adjust the interest rate r .]

Continued...

2. (25 marks)

- (a) Consider a model with three periods and one risky security where the interest rate is $r = 0.1$ and the price process is given by the following table:

State	$t = 0$	$t = 1$	$t = 2$	$t = 3$
ω_1	10	16	20	24
ω_2	10	16	20	16
ω_3	10	16	18	20
ω_4	10	16	18	16
ω_5	10	8	12	16
ω_6	10	8	12	10
ω_7	10	8	6	8
ω_8	10	8	6	4

- i. **(3 marks)** What does it mean for a random variable τ to be a stopping time with respect to a filtration $\{\mathcal{F}_t\}$?
State the definition of the Snell envelope Z_s of a stochastic process $\{X_t, t = 0, \dots, T\}$ adapted to some filtration $\{\mathcal{F}_t\}$.
State the optimal stopping rule at any time t , in terms of Z and X .
 - ii. **(2 marks)** Determine the payoff process of an American put option with strike price $K = 10$ on one share of the stock.
 - iii. **(7 marks)** Find the time 0 price for the American put option considered in ii. Determine an optimal stopping time and find a replicating portfolio for the trading strategy that uses this optimal stopping time.
- (b) Consider a Cox-Ross-Rubinstein model with parameters u and d and $T = 2$ periods where $u = 1/d$, $r = 0$ and $S_0 = 1$.
- i. **(3 marks)** Associate the four possible stock price movements with states $\omega_1, \dots, \omega_4$ and calculate $S_t(\omega)$ for $t = 1, 2$ and each ω in terms of d .
 - ii. **(2 marks)** Find the risk neutral probability measure for one period of this model in terms of d .
 - iii. **(2 marks)** Calculate $c(K)$, the time $t = 0$ value of the European call with strike price K for $K = 1$ and for $K = d^2$.
 - iv. **(6 marks)** Calculate the time $t = 0$ value of the look-back call option with average strike, i.e. the contingent claim expiring at $t = 2$ with pay off $(S_2 - K)^+$ where

$$K = K_{av}(\omega) = \frac{1}{3} \sum_{t=0}^2 S_t(\omega).$$

Verify that the look-back is worthless in exactly the two states ω_2, ω_4 and its value is $\frac{2}{3}c(1)$.

Continued...

3. (25 marks)

(a) Let $\{X_t, t \geq 0\}$ be a simple random walk with parameter p .

- i. (2 marks) What is the distribution of $Y_t = X_t - X_{t-1}$? For which values of p is (X_t) a martingale and for which a submartingale?
- ii. (5 marks) Consider the process $\{L_t, t \geq 0\}$ defined by $L_0 = \alpha$ and

$$L_t = \beta^{X_t} \quad (t \geq 1).$$

Find all values for α and β such that $\{L_t, t \geq 0\}$ is a martingale and prove your claim.

(b) Let $(W_t)_{t \geq 0}$ be standard Brownian motion.

- i. (2 marks) State the properties satisfied by the distribution of the increments of W_t
- ii. (2 marks) Calculate the probability that $W_{10} \geq 5$.
- iii. (2 marks) Show that for any $s, t \geq 0$ the covariance satisfies $\text{Cov}(W_s, W_t) = \min(s, t)$.
- iv. (2 marks) Define a new process by $B_0 = 0$ and $B_t = tW_{1/t}$ for $t > 0$. Show that $\text{Cov}(B_s, B_t) = \min(s, t)$.

(c) (4 marks) Consider a stock process of the form

$$S_t = \exp(\sigma W_t + (\mu - \sigma^2/2)t),$$

and a bond process

$$B_t = \exp(rt).$$

For any $t \geq 0$, find the probability that the stock is more valuable than the bond at time t . Find the limit of this probability as $t \rightarrow 0$.

- (d) (3 marks) Write down Itô's formula for the function $f(t, W_t)$ of a standard Brownian motion W_t , where $f = f(t, x)$ is a function of both t and x . Then use Itô's formula to put the process $X_t = \exp(5W_t)$ in standard form $dX_t = \mu_t dt + \sigma_t dW_t$. How can you see in this representation, whether or not X_t is a martingale?
- (e) (3 marks) Explain what is meant by a simple stochastic process X , and give a formula for the stochastic integral $I_t(X)$ of a simple stochastic process.

Continued...

4. (25 marks)

- (a) Consider the Black-Scholes formula for the value of a certain European option, given as:

$$\Pi_t = S_t \Phi(d_1(S_t, T - t)) - Ke^{-r(T-t)} \Phi(d_2(S_t, T - t)) \text{ for } 0 \leq t \leq T.$$

The functions $d_1(u, v)$ and $d_2(u, v)$ are given by

$$d_1(u, v) = \frac{\log(u/K) + (r + \sigma^2/2)v}{\sigma\sqrt{v}} \quad \text{and} \quad d_2(u, v) = \frac{\log(u/K) + (r - \sigma^2/2)v}{\sigma\sqrt{v}}.$$

- i. **(5 marks)** What do the terms t , T , r , Φ and σ represent?
In the case $S_t = 10$, $t = 0$, $T = 10$, $r = 0.1$, $K = 13$, $\sigma = 1$, calculate the value of Π_t .
 - ii. **(2 marks)** The formula is based on a model containing only one risky stock of value S_t and one riskless bond. At time t , how much of each should a trader wishing to replicate the option hold?
 - iii. **(2 marks)** What is the time 0 price of a European put option with the same strike price as in i.?
 - iv. **(5 marks)** Calculate the limit of Π_t as $t \rightarrow T$, keeping all other terms constant. State the formula for the claim that this formula prices. What would you expect the limit of Π_t to be as $K \rightarrow \infty$, keeping all other terms constant?
- (b) Let S_t be the price of an equity that satisfies the Black-Scholes model $dS_t = \mu S_t dt + \sigma S_t dW_t$ for constants $\mu \in \mathbb{R}$, $\sigma > 0$ and where $(W_t)_{t \geq 0}$ is the standard Brownian motion. Assume that the risk-free rate of interest is r .
- i. **(6 marks)** A binary Cash-or-Nothing Call yields a payoff of C at time T if $S_T \geq K$ and 0 otherwise. Find its arbitrage-free price at time $t = 0$.
 - ii. **(5 marks)** Find all financial instruments based on the Black-Scholes model such that their arbitrage-free price $P(t, S_t)$ satisfies the Black-Scholes-Merton equation but is deterministic. Give a financial interpretation of your results.

End of examination.

Probabilities of a standard normal distribution

The next tables shows probabilities from a standard normal distribution. Each cell contains the value of $\Phi(x_1 + x_2)$ where x_1 and x_2 are the labels of the corresponding row and column.

Φ	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990